



Databases and its Applications

Relational Modelling

Pooja T S
Computer Applications

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Mapping ER/EER to Relational Schema

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Introduction – ER to Relational Mapping

► Purpose of Mapping:

- Convert conceptual ER design into logical relational schema.
- Bridge between **high-level ER model** and **database implementation**.
- Ensures all entities, attributes and relationships are represented in tabular form.

- **Goal:** Preserve semantics and constraints of the ER diagram during mapping.



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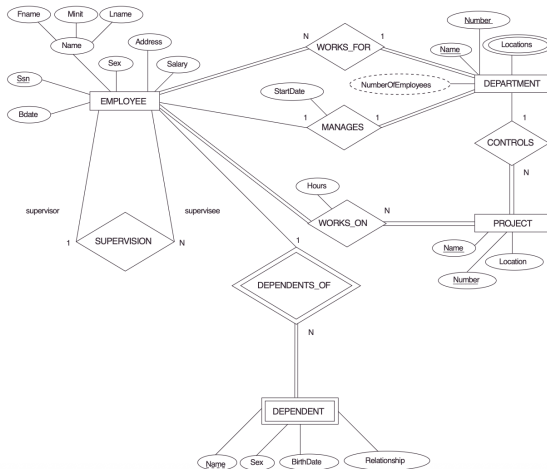
Overall Mapping Process

- ▶ The ER-Relational mapping follows these standard steps:
 1. Mapping of Regular (Strong) Entity Types
 2. Mapping of Weak Entity Types
 3. Mapping of Binary 1:1 Relationship Types
 4. Mapping of Binary 1:N Relationship Types
 5. Mapping of Binary M:N Relationship Types
 6. Mapping of Multivalued Attributes
 7. Mapping of N-ary Relationship Types
- ▶ Each rule converts ER components into relational tables.



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Company Database ER Diagram





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ER/EER to Relational Schema Mapping



- ▶ We now apply the ER-Relational Mapping Algorithm to the **Company Database ER Diagram**.
- ▶ The ER model includes:
 - Entities – EMPLOYEE, DEPARTMENT, PROJECT, DEPENDENT
 - Relationships – WORKS_FOR, MANAGES, CONTROLS, WORKS_ON
 - Attributes – including composite, multivalued, and derived types
- ▶ Each step of the algorithm is demonstrated on this schema.



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Step 1 – Mapping Strong Entity Types

- ▶ **Rule 1:** Create a relation for each strong entity type.
 - Include all simple attributes.
 - Choose the key attribute as the **Primary Key**.
- ▶ **Entities in Company Schema:**
 - EMPLOYEE(SSN, FName, Minit, LName, BDate, Address, Sex, Salary, SuperSSN, DNo)
 - DEPARTMENT(DNumber, DName, MgrSSN, MgrStartDate)
 - PROJECT(PNumber, PName, PLocation, DNum)
- ▶ Primary Keys: SSN, DNumber, PNumber



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Step 1 – Mapping Strong Entity Types

EMPLOYEE

<u>Fname</u>	Minit	<u>Lname</u>	Ssn	Bdate	Address	Sex	Salary
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DEPARTMENT

<u>Dname</u>	<u>Dnumber</u>
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PROJECT

Pname	<u>Pnumber</u>	Plocation
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Notes – Strong Entity Mapping

► Foreign Keys introduced later:

- SuperSSN → Supervisor relation (1:1)
- DNo, DNum → DEPARTMENT relation (1:N)

► Composite attributes:

- Name = (FName, Minit, LName) flattened into three attributes.



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Step 2 – Mapping Weak Entity Types

► **Rule 2:** For each weak entity:

- Create a relation including its partial key and attributes.
- Add the primary key of the owner entity as a **Foreign Key**.
- Primary key = (Owner PK + Partial Key).

► **Example:**

- DEPENDENT(ESSN, DependentName, Sex, BDate, Relationship)
- PK = (ESSN, DependentName)
- FK = ESSN references EMPLOYEE(SSN).

► **Image suggestion:** Weak entity DEPENDENT attached to EMPLOYEE.



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Step 2 – Mapping Weak Entity Types

DEPENDENT

<i>Essn</i>	<u>Dependent_name</u>	Sex	Bdate	Relationship
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- ▶ **Rule 3:** Choose one participating entity — preferably the one with total participation.
 - Include the primary key of the other entity as a foreign key.
- ▶ **Example: MANAGES**
 - Each DEPARTMENT is managed by one EMPLOYEE.
 - Add MgrSSN to DEPARTMENT.
 - PK = DNumber; FK = MgrSSN references EMPLOYEE (SSN).



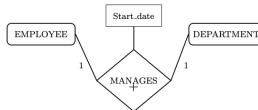
Step 3 – Mapping Binary 1:1 Relationships

DEPARTMENT

Dname	<u>Dnumber</u>
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DEPARTMENT

Dname	<u>Dnumber</u>	Mgr_ssn	Mgr_start_date
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- ▶ **Rule 4:** Add the primary key of the “one” side as a **foreign key** in the “many” side.
- ▶ **Example 1 – WORKS_FOR**
 - Each employee works for one department.
 - Add DNo in EMPLOYEE.
 - PK = SSN; FK = DNo references DEPARTMENT (DNumber).
- ▶ **Example 2 – CONTROLS**
 - Each department controls several projects.
 - Add DNum in PROJECT.
 - PK = PNumber; FK = DNum references DEPARTMENT (DNumber).



Step 4 – Mapping Binary 1:N Relationships

EMPLOYEE

Fname	Minit	Lname	<u>Ssn</u>	Bdate	Address	Sex	Salary
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EMPLOYEE

Fname	Minit	Lname	<u>Ssn</u>	Bdate	Address	Sex	Salary	Super_ssn	Dno
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PROJECT

Pname	<u>Pnumber</u>	Plocation
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PROJECT

Pname	<u>Pnumber</u>	Plocation	Dnum
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Step 5 – Mapping Binary M:N Relationships

- ▶ **Rule 5:** Create a new relation for each M:N relationship.
 - Include the PKs of both participating entities as **Foreign Keys**.
 - Their combination forms the **composite Primary Key**.
- ▶ **Example – WORKS_ON**
 - WORKS_ON (ESSN, PNo, Hours)
 - PK = (ESSN, PNo)
 - FK1 = ESSN → EMPLOYEE (SSN)
 - FK2 = PNo → PROJECT (PNumber).



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Step 5 – Mapping Binary M:N Relationships

EMPLOYEE

Fname	Minit	Lname	<u>Ssn</u>	Bdate	Address	Sex	Salary	Super_ssn	Dno
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PROJECT

Pname	<u>Pnumber</u>	Plocation	Dnum
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WORKS_ON

<u>Essn</u>	<u>Pno</u>	Hours
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Here, the many-to-many relationship between EMPLOYEE and PROJECT is mapped as:

- Each combination of Essn (Employee) and Pno (Project) forms a composite primary key.
- Both Essn and Pno act as foreign keys referencing **EMPLOYEE(Ssn)** and **PROJECT(Pnumber)**.
- The relationship attribute **Hours** is included as a regular attribute in the WORKS_ON table.



► **Rule 6:** Create a separate relation for each multivalued attribute.

- Include the attribute and the PK of the entity it belongs to.
- The combination of both is the **Primary Key**.

► **Example – DEPT_LOCATIONS**

- DEPT_LOCATIONS(DNumber, DLocation)
- PK = (DNumber, DLocation)
- FK = DNumber references DEPARTMENT(DNumber).



DEPT_LOCATIONS

<u>Dnumber</u>	<u>Dlocation</u>
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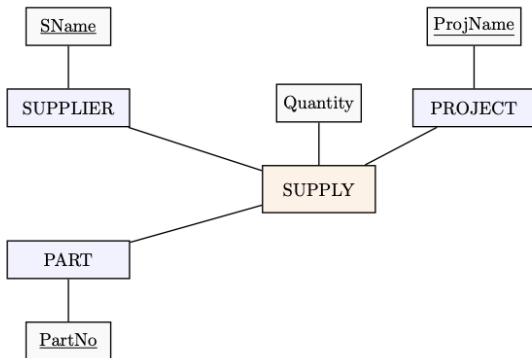
Step 7 – Mapping N-ary Relationships

- ▶ **Rule 7:** For relationships involving 3 or more entities:
 - Create a new relation containing the PKs of all participant entities as **Foreign Keys**.
 - Their combination forms the composite **Primary Key**.
 - Add any descriptive attributes.
- ▶ **Image suggestion:** Example N-ary relationship such as SUPPLY(Part, Supplier, Project).



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Step 7 – Mapping N-ary Relationships





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Step 7 – Mapping N-ary Relationships

SUPPLIER

<u>SName</u>	...
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PROJECT

<u>ProjName</u>	...
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PART

<u>PartNo</u>	...
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SUPPLY

<u>SName</u>	<u>ProjName</u>	<u>PartNo</u>	Quantity
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Resulting Relational Schema

- ▶ **EMPLOYEE**(SSN, FName, Minit, LName, BDate, Address, Sex, Salary, SuperSSN, DNo)
- ▶ **DEPARTMENT**(DNumber, DName, MgrSSN, MgrStartDate)
- ▶ **PROJECT**(PNumber, PName, PLocation, DNum)
- ▶ **WORKS_ON**(ESSN, PNo, Hours)
- ▶ **DEPENDENT**(ESSN, DependentName, Sex, BDate, Relationship)
- ▶ **DEPT_LOCATIONS**(DNumber, DLocation)



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Summary – Applied Mapping Concepts



- ▶ All entities, relationships and attributes of the Company ER diagram are now mapped to relations.
- ▶ Relationships represented via foreign keys ensure **referential integrity**.
- ▶ Weak entities use **composite primary keys**.
- ▶ Multivalued attributes are stored in separate relations.
- ▶ **Image suggestion**: Side-by-side comparison – ER diagram and final relational schema.



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Summary

Correspondence between ER and Relational Models

ER Model	Relational Model
Entity type	"Entity" relation
1:1 or 1:N relationship type	Foreign key (or "relationship" relation)
M:N relationship type	"Relationship" relation and two foreign keys
n-ary relationship type	"Relationship" relation and n foreign keys
Simple attribute	Attribute
Composite attribute	Set of simple component attributes
Multivalued attribute	Relation and foreign key
Value set	Domain
Key attribute	Primary (or secondary) key



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CELEBRATING 50 YEARS

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